

Perfume Business AI Prediction & Advisory System

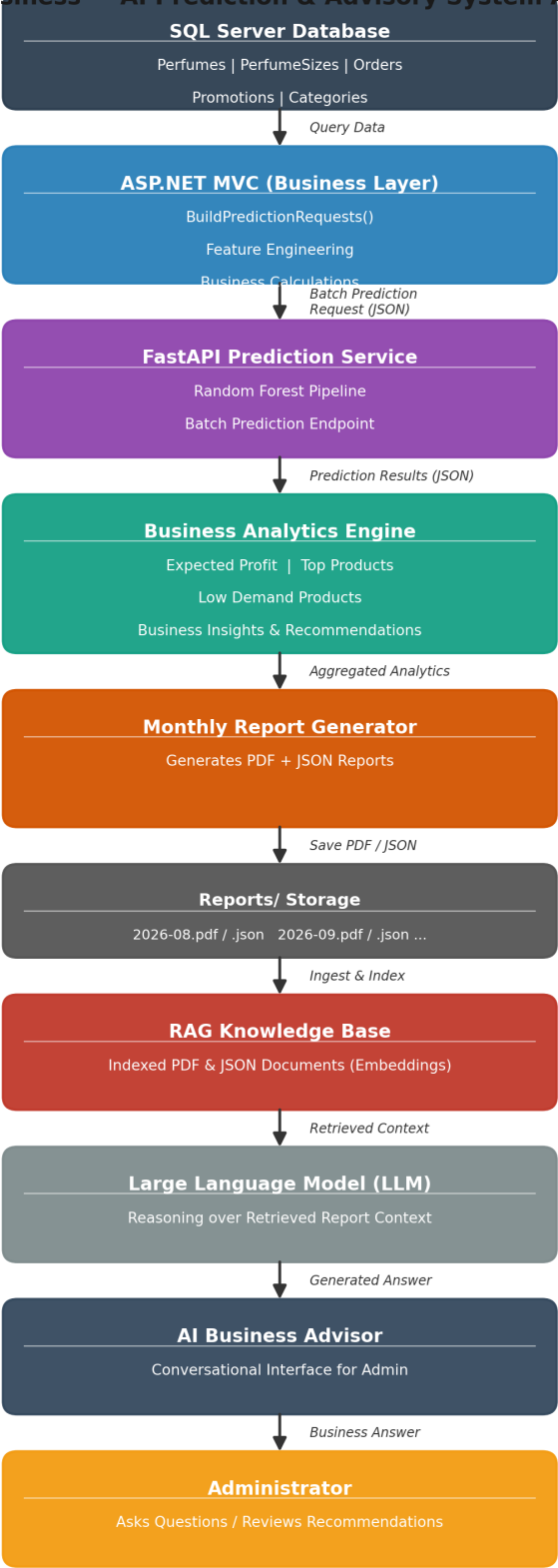
Architecture Diagram & Workflow Diagram

This document presents the end-to-end system architecture — from the SQL Server data layer through feature engineering, machine learning prediction, business analytics, report generation, and the RAG-powered AI Business Advisor — along with the step-by-step workflow followed when an administrator asks the AI advisor a business question.

1. System Architecture

The diagram below shows how data flows from the SQL Server database, through the ASP.NET MVC business layer (feature engineering and business calculations), into the FastAPI batch prediction service, and onward through analytics, reporting, and the RAG knowledge base into the LLM-powered AI Business Advisor.

Perfume Business — AI Prediction & Advisory System Architecture



2. Admin Q&A; Workflow

The diagram below illustrates the runtime workflow that occurs when an administrator asks the AI Business Advisor a question: the question is embedded and matched against the RAG knowledge base, relevant report excerpts are retrieved, the LLM analyzes the prediction results and analytics context, and a grounded recommendation is returned to the administrator.

Admin Q&A Workflow — AI Business Advisor

